

AI Healthcare Assistant using Hugging Face & Prompt Engineering

Introduction

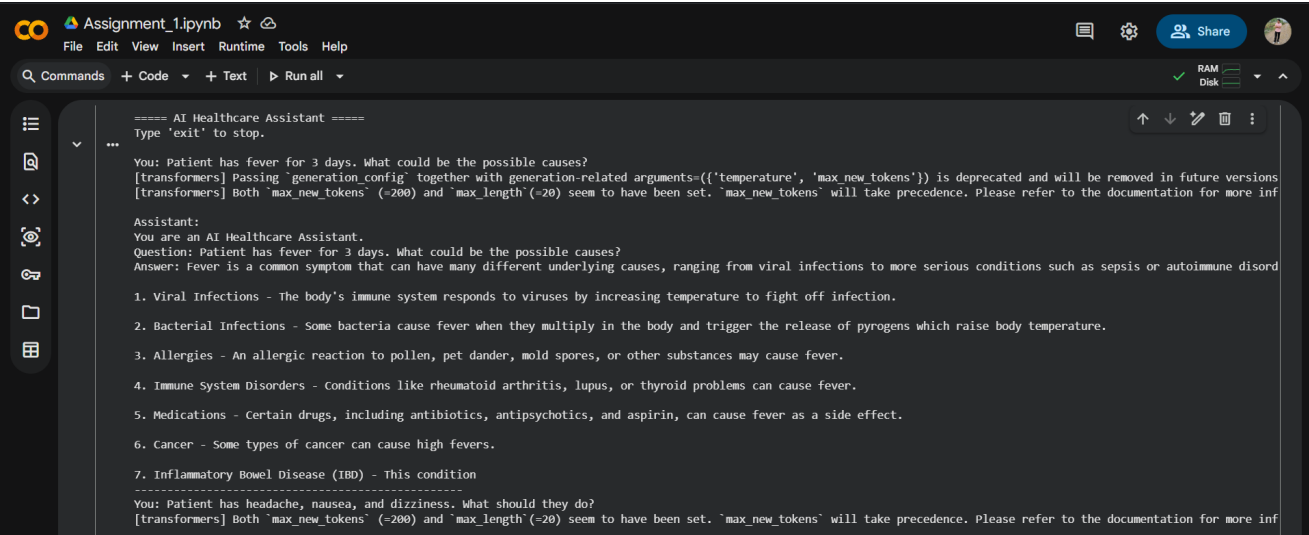
The aim of the project was to create an AI Healthcare Assistant employing Hugging Face Transformers library and utilizing various Prompt Engineering methods in order to enhance the quality of the answers. The chatbot was implemented in a way to respond to queries regarding healthcare, provide structured information, minimize the hallucinations, and customize its responses depending on a particular user's role. Google Colab was used to implement the different tasks.

Model Selection

In this project, the choice fell on Qwen2.5-1.5B-Instruct model from the Hugging Face library. This is an instruction adjusted lightweight model which works well in Google Colab environment. This model demonstrates decent potential in terms of following and responding to instructions.

Task 1 – Hugging Face Chatbot

This task was devoted to creating an interactive AI Healthcare chatbot which was developed by means of the Hugging Face Transformers library. The chatbot was created using AutoTokenizer, AutoModelForCausalLM as well as implementing pipeline() API. Users had the opportunity to enter various health-related requests and receive adequate educational replies.



```
===== AI Healthcare Assistant =====
Type 'exit' to stop.

...
You: Patient has fever for 3 days. What could be the possible causes?
[transformers] Passing 'generation_config' together with generation-related arguments-({'temperature', 'max_new_tokens'}) is deprecated and will be removed in future versions
[transformers] Both 'max_new_tokens' (~200) and 'max_length' (~20) seem to have been set. 'max_new_tokens' will take precedence. Please refer to the documentation for more info

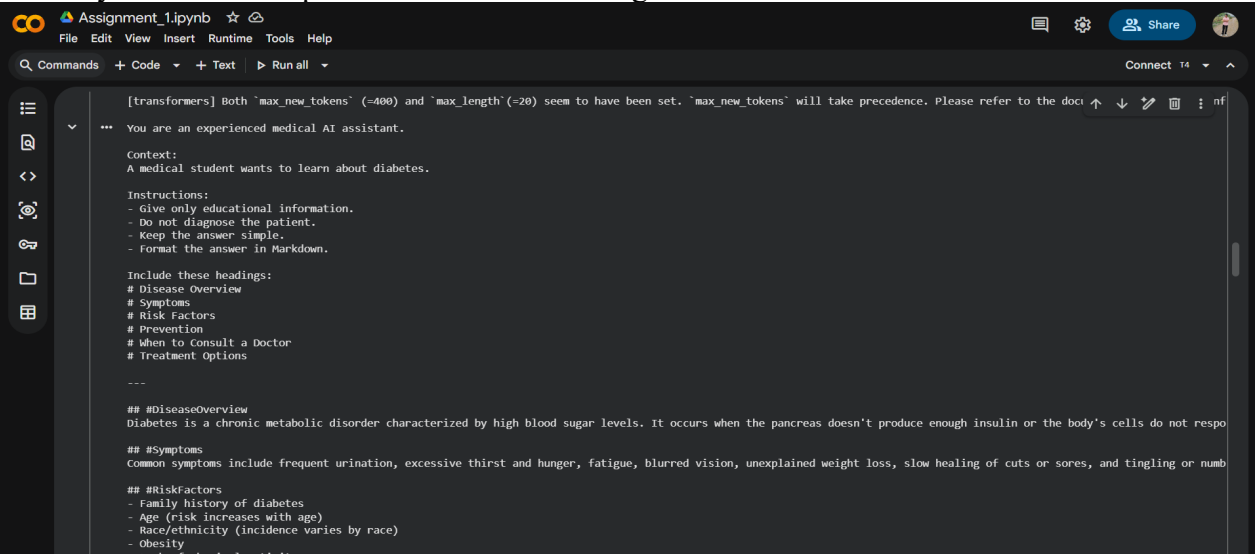
Assistant:
You are an AI Healthcare Assistant.
Question: Patient has fever for 3 days. What could be the possible causes?
Answer: Fever is a common symptom that can have many different underlying causes, ranging from viral infections to more serious conditions such as sepsis or autoimmune disorders.

1. Viral Infections - The body's immune system responds to viruses by increasing temperature to fight off infection.
2. Bacterial Infections - Some bacteria cause fever when they multiply in the body and trigger the release of pyrogens which raise body temperature.
3. Allergies - An allergic reaction to pollen, pet dander, mold spores, or other substances may cause fever.
4. Immune System Disorders - Conditions like rheumatoid arthritis, lupus, or thyroid problems can cause fever.
5. Medications - Certain drugs, including antibiotics, antipsychotics, and aspirin, can cause fever as a side effect.
6. Cancer - Some types of cancer can cause high fevers.
7. Inflammatory Bowel Disease (IBD) - This condition

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You: Patient has headache, nausea, and dizziness. What should they do?
[transformers] Both 'max_new_tokens' (~200) and 'max_length' (~20) seem to have been set. 'max_new_tokens' will take precedence. Please refer to the documentation for more info
```

Task 2 – Prompt Engineering

The prompt "Tell me about diabetes" was used to implement Prompt Engineering principles. Techniques like Role Prompting, Context, Constraints, and Output Formatting were employed to obtain a structured response in Markdown format containing sections: Disease Overview, Symptoms, Risk Factors, Prevention, and When to Contact the Physician. The output became clearer and organized.



```
[transformers] Both 'max_new_tokens' (~400) and 'max_length' (~20) seem to have been set. 'max_new_tokens' will take precedence. Please refer to the documentation for more info

...
You are an experienced medical AI assistant.

Context:
A medical student wants to learn about diabetes.

Instructions:
- Give only educational information.
- Do not diagnose the patient.
- Keep the answer simple.
- Format the answer in markdown.

Include these headings:
# Disease Overview
# Symptoms
# Risk Factors
# Prevention
# When to Consult a Doctor
# Treatment Options

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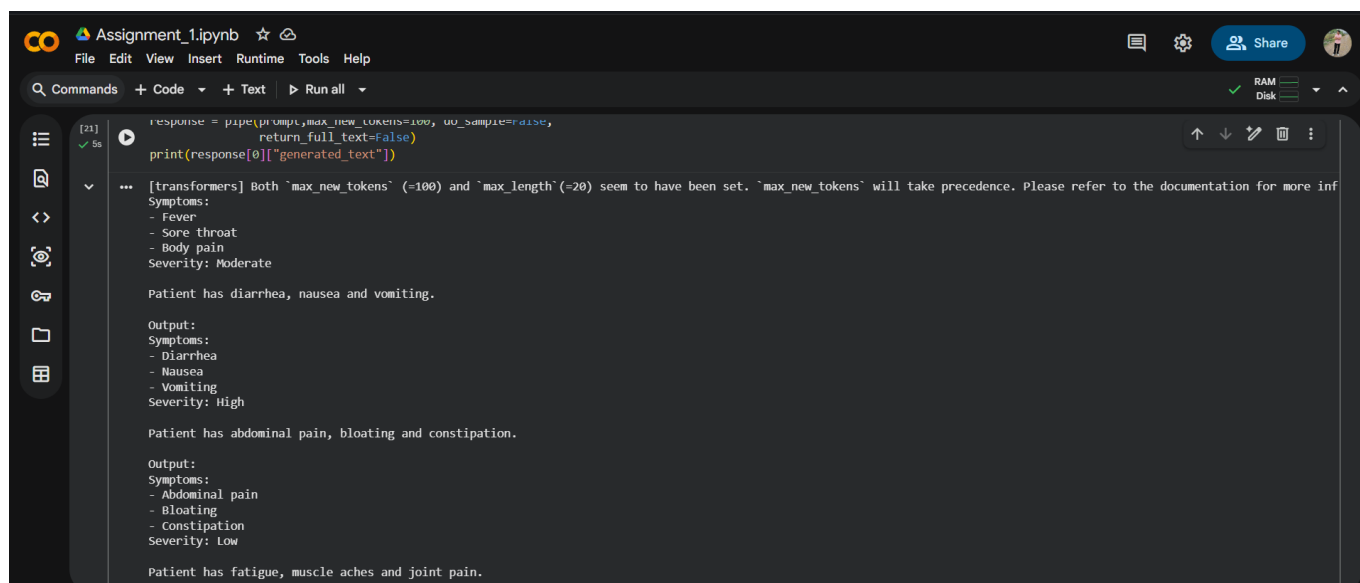
## #DiseaseOverview
Diabetes is a chronic metabolic disorder characterized by high blood sugar levels. It occurs when the pancreas doesn't produce enough insulin or the body's cells do not respond properly to insulin.

## #Symptoms
Common symptoms include frequent urination, excessive thirst and hunger, fatigue, blurred vision, unexplained weight loss, slow healing of cuts or sores, and tingling or numbness in the hands and feet.

## #RiskFactors
- Family history of diabetes
- Age (risk increases with age)
- Race/ethnicity (incidence varies by race)
- Obesity
```

Task 3 – Few-Shot Prompting

Few-Shot Prompting was used to provide three examples of clinical documentation with the structured summaries before the final prompt. The technique allowed the algorithm of AI to produce consistent outputs according to the pattern used in the previous responses.



```
[21] response = pipe(prompt, max_new_tokens=100, do_sample=False,
55             return_full_text=False)
print(response[0]["generated_text"])

...

[transformers] Both `max_new_tokens` (=100) and `max_length` (=20) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more info.

Symptoms:
- Fever
- Sore throat
- Body pain
Severity: Moderate

Patient has diarrhea, nausea and vomiting.

Output:
Symptoms:
- Diarrhea
- Nausea
- Vomiting
Severity: High

Patient has abdominal pain, bloating and constipation.

Output:
Symptoms:
- Abdominal pain
- Bloating
- Constipation
Severity: Low

Patient has fatigue, muscle aches and joint pain.
```

Task 4 – Chain of Thought Prompting

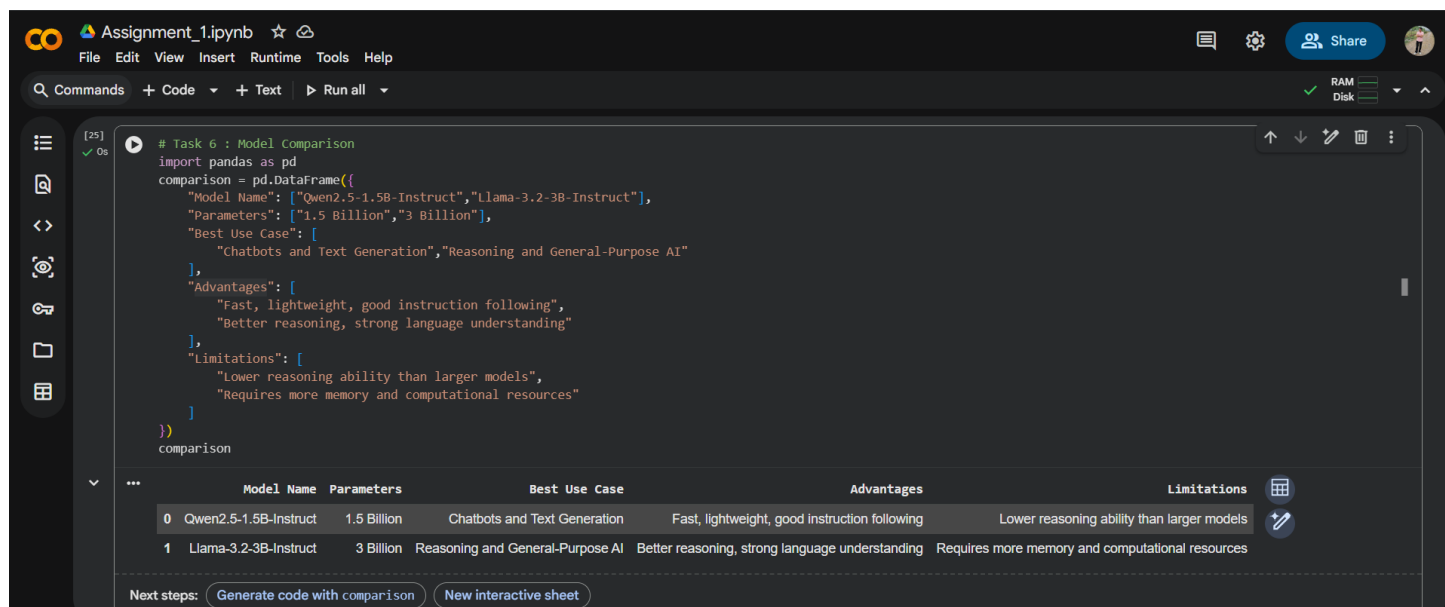
A Chain of Thought prompt was constructed for the purpose of being able to analyze the symptoms presented by the patient and then provide the recommendation. In this case, the prompt was designed to keep the internal reasoning of the model confidential, but provide the logic behind the answer.

Task 5 – Structured Output

The model was prompted to respond in a certain JSON format including disease, symptoms, risk level, and recommendation. A structured output helps to facilitate information usage.

Task 6 – Model Comparison

Two Hugging Face instruction-tuning models were compared considering parameters, scenarios of their best usage, advantages, and downsides. Comparison allows awareness of various strengths and weaknesses of the language models for healthcare.



```
[25] # Task 6 : Model Comparison
import pandas as pd
comparison = pd.DataFrame({
    "Model Name": ["Qwen2.5-1.5B-Instruct", "Llama-3.2-3B-Instruct"],
    "Parameters": ["1.5 Billion", "3 Billion"],
    "Best Use Case": [
        "Chatbots and Text Generation", "Reasoning and General-Purpose AI"
    ],
    "Advantages": [
        "Fast, lightweight, good instruction following",
        "Better reasoning, strong language understanding"
    ],
    "Limitations": [
        "Lower reasoning ability than larger models",
        "Requires more memory and computational resources"
    ]
})
comparison
```

	Model Name	Parameters	Best Use Case	Advantages	Limitations
0	Qwen2.5-1.5B-Instruct	1.5 Billion	Chatbots and Text Generation	Fast, lightweight, good instruction following	Lower reasoning ability than larger models
1	Llama-3.2-3B-Instruct	3 Billion	Reasoning and General-Purpose AI	Better reasoning, strong language understanding	Requires more memory and computational resources

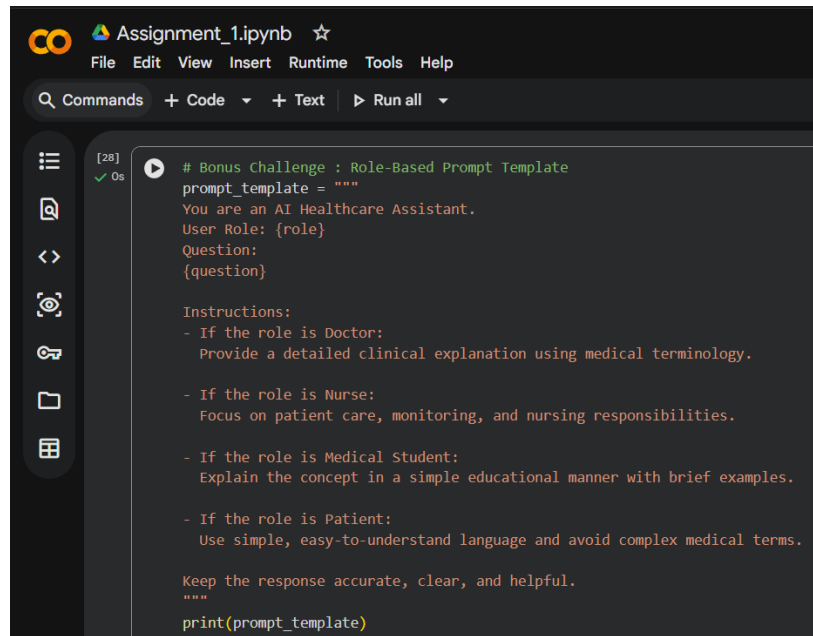
Next steps: [Generate code with comparison](#) [New interactive sheet](#)

Task 7 – Prompt Optimization

An unsatisfactory response was analyzed for issues including hallucinations, unclear answers, etc. After that prompt engineering techniques such as role prompting, constraints, context, and structured output were used to modify and improve the prompt.

Bonus Challenge – Role-Based Prompt Template

A new prompt template has been developed in a dynamic way which enables chatbots to give different responses based on users' characters. This is possible due to Hugging Face ML model as a prompt template helps to choose responses relevant to either Doctor, Nurse, Medical student, or Patient.



```
[28] ✓ 0s
# Bonus Challenge : Role-Based Prompt Template
prompt_template = """
You are an AI Healthcare Assistant.
User Role: {role}
Question:
{question}

Instructions:
- If the role is Doctor:
  Provide a detailed clinical explanation using medical terminology.

- If the role is Nurse:
  Focus on patient care, monitoring, and nursing responsibilities.

- If the role is Medical Student:
  Explain the concept in a simple educational manner with brief examples.

- If the role is Patient:
  Use simple, easy-to-understand language and avoid complex medical terms.

Keep the response accurate, clear, and helpful.
"""
print(prompt_template)
```

Challenges Faced

During the project realization, a lot of problems appeared. For example, it sometimes occurred that the output from the model was longer than expected, especially during Few-Shot Prompting. Hugging Face warned regarding the parameters of generating text but it did not affect the chatbot. . Creating a structure prompt that allows reducing hallucinations is a challenging process that requires a lot of improvements.

Improvements After Prompt Refinement

Prompt refinement has had a great impact on the performance of chatbots. The formulated prompts have made the responses more well-organized, informative, and consistent. The methodologies of Role Prompting, Constraints, Few-Shot Prompting, Structured Output, and Role-based Prompting have minimized the vagueness of prompts and led to the generation of more accurate responses. The chatbot was capable of producing the required responses depending on the user type without any modifications of the original language model used by the chatbot.

Conclusion

The success of the project has proved using Hugging Face Transformers and Prompt Engineering methods to build an AI healthcare assistant. The use of various prompt techniques improved prompts, reduced the hallucinations, and led to generating structured and role-specific outputs. The project emphasizes the significance of proper prompt creation in treatment of large language models for healthcare purposes.